

UNIT-3

Risk and Vulnerability

Risk and Vulnerability: Building codes and land use planning-Social Vulnerability

Environmental vulnerability Macro-economic management and sustainable development.

Climate change risk rendition financial management of disaster related losses

Building codes and land use planning:

Building codes:

- A building code is a set of regulations written by city or county officials with the help of construction pros that governs the design, construction and modification of commercial buildings, homes and other structures in the jurisdiction.
- Violating these regulations can result in fines and/or penalties.

National Building Code (NBC):

- The National Building Code of India (NBC) is an instrument issued by the Bureau of Indian Standards (BIS).
- **NBC** provides guidelines to the building construction activities all over the nation. It has given distinct building codes for residential, institutional, commercial, industrial, schools, colleges, etc. It is crucial to abide by all these guidelines, even for all government agencies, private construction companies, public work departments, etc.
- **The Bureau of Indian Standards** is the National Standard of India. The smooth development standardization, marking, and quality certification of goods and related things is under the purview of BIS.

Types of buildings as per NBC:

Residential, educational, institutional, assembly buildings, industrial buildings, storage buildings, etc., are the types of buildings as per NBC.

Types of industrial buildings as per NBC:

- Buildings for low-hazardous industry
- Facilities for medium hazardous industry
- Structures for highly hazardous industries

National Building Code of India – General Requirements for Residential Buildings:

Kitchen, bathroom, basement, garage, open spaces (within a plot), area and height limitations, Off street parking spaces, staircase/exit requirements, fire and life safety

Land use planning:

The Role of land use planning in disaster risk reduction

INTRODUCTION

- 85 percent of people exposed to earthquakes, cyclones, floods and droughts live in developing countries.
- The massive cost of disasters poses a significant threat to achieving the development goals. Direct disaster damage costs alone have shot up from US\$ 75.5 billion in the 1960s to nearly a trillion dollars in the past ten years.
- Even these ballooning figures cannot capture the long-term cost to the people and communities that bear the brunt of disasters, paying with their lives, their livelihoods and their future development prospects.
- Land use planning is a newly emerging disaster reduction method.
- Land use planning is potentially powerful mitigation tool as it seeks to mitigate the risks and vulnerability from several geological and other hazards.
- So, the logic of land use planning is compatible with disaster risk reduction because both are systematic, future-oriented, decision-oriented, and proactive.

Urban Land Use and Significance of Planning

The planning and control of land use includes a vast spectrum of general policies about it which leave determining effects and influences on programs and plans of urban development and cover economic, environmental, social and political aspects.

Land use planning in the present cities includes two main categories:

- The expansion of the city,
- City restoration,

The degree of effectiveness of land use planning and its efficiency in case of an earthquake from the time of the earthquake till the return of the city to normal conditions is significant and determines two indicators of vulnerability (Casualties and Destruction).

- Due to structural issues, in Iran, land use plans have been restricted to land use maps, per capita use tables, and regulations of zoning.
- The most important criteria for determining appropriate locations for different urban activities and urban land use can be considered as the followings:
 1. Compatibility,
 2. Comfort,
 3. Efficiency,
 4. Favourability,
 5. Health and
 6. Safety.

- Attending to these factors and considering them in the land use planning improves efficiency of the city, welfare and comfort for the citizens, and minimizing the fatalities, and economical damages because of the earthquake.

Organizing the urban structure

- Consideration of land use features such as distribution, adjacency, compatibility with other urban elements and spaces
- Creating hierarchy in urban open spaces
- Establishing an efficient communication network against earthquake
- Observing the high-risk areas borders and preventing urban sprawl towards them
- It is quite axiomatic that recognition of different quakes, lands prone to breakage of crust, landslides, land sinking, breakage of surface faults, and observing the related standards is essential in urban land use planning.
- In the following table, the geological factors and the consequences of earthquake phenomenon have been categorized into four sites with their features and characteristics.

Organizing the urban structure

Site	Risk Level	Characteristics
A	No Risk	▪ Low probability of earth subsidence ▪ Correlation between soil grains ▪ Existence of Neon and Quaternary layers
B	Low Risk	▪ Low subsidence due to the movement of confined fine-grained layer in alluvial
C	Moderate Risk	▪ Moderate subsidence due to the sturdy soft clay ▪ Having liquefaction potential ▪ Low probability of occurrence of rock fall ▪ Sites located at distances up to 300- 500 m from faults
D	High Risk	▪ Significant Depression or Different subsidence and weak surface-soil due to Earth's layers depression because of earthquake waves ▪ Geological displacement effects ▪ Earth Severe shock because of Earthquake waves ▪ Having landslides potential ▪ Having rock-fall potential ▪ Having liquefaction potential

urban fabric rehabilitation

➤ Aggregation of land lots:

The problems and issues related to urban districts and lots in inefficient urban fabrics can be categorized into the following groups:

The pieces of land have a small area, in a way that considering the least open space, that is the yard, one cannot construct an efficient building in that, in other words, the land is wasted.

The piece of land does not enjoy proper dimensions, particularly the façade dimension.

More importantly, the lot is not capable of correcting the route and there would be unused space.

The lots have inappropriate conditions with regard to topographic, and position issues, and inappropriate length makes access difficult for them.

►Renovation of Old Textures:

Among important factor which exacerbate the destruction in an earthquake is the old land uses. These need to be recognized and renovated.

conclusion

- The most important problem in land use planning for disaster risk reduction is the determination of the appropriate criteria.
- For the determination of a range for the relevant criteria, identification of all effective criteria such as social, cultural, geographical, political and economic criteria are necessary.
- Lack of attention to any of these criteria may lead to the failure of the planning. In this research the necessary criteria for the promotion of quantitative and qualitative land-use planning to confront the negative effects of natural disasters are presented.

Social Vulnerability:

- Social vulnerability refers to potential harm to people. It involves a combination of factors that determine the degree to which someone's life and livelihood are put at risk by a discrete and identifiable event in nature or in society.
- Social vulnerability refers to the characteristics of a person or group in terms of their capacity to anticipate, cope with, resist and recovery from the impact of a natural hazard (Wisner et al.,2004)."
- Social vulnerability refers to the resilience of communities when confronted by external stresses on human health, stresses such as natural or human-caused disasters, or disease outbreaks. Reducing social vulnerability can decrease both human suffering and economic loss
- Social vulnerability refers to the inability of people, organizations, and societies to withstand
- adverse impacts from multiple stressors to which they are exposed. These impacts are due in part to characteristics inherent in social interactions, institutions, and systems of cultural values
- Social vulnerability refers to the susceptibility of social groups to potential losses from hazard events of society's resistance and resilience to hazard

The goals of this project are to:

1. Identify social vulnerability factors that have a major impact on the response of communities to natural disasters.
2. Develop a social vulnerability index that can predict communities that may struggle in the aftermath of a natural disaster.
3. Apply the model to three (3) past natural disasters and validate the outcome based on available historical. data.

Successful completion of this project can provide an insight into the vulnerabilities of communities against natural disasters and offer a framework for targeted investments to ensure resilience.

Climate change has led to an increase in the frequency and severity of natural disasters, which further signifies the importance of this problem.

Environmental Vulnerability:

The concept of vulnerability emphasizes the social characteristics and configurations used by communities to face the challenges of the physical environment.

It refers to both, the susceptibility to the negative socioeconomic impacts of environmental variability and the degree to which a community is capable of coping with, resisting, and recovering from the impacts of specific environmental events.

Healthy, productive and protective environments, social systems and economies are the basis of sustainable development and human welfare.

The environment is the source of all our raw materials and absorbs the pollution from our activities.

In turn, whilst going about our daily business (social and economic) we use the environment and convert its resources and natural services into those that directly support us.

The problem is that all of these systems can be damaged, overloaded, or prevented from meeting our needs.

By our own choices we can to a large extent determine our own quality of life, the condition of our lands and opportunities for future generations.

Macroeconomic Management and Sustainable development:

Macroeconomics is a branch of economics dealing with the performance, structure, behaviour, and decision-making of an economy as a whole. This includes regional, national, and global economies.

The five main objectives or goals of macroeconomics are:

- Stable and sustainable economic growth.
- Low levels of inflation.
- Low rates of unemployment.
- Equitable distribution of income in a country.
- There should be an equilibrium in the balance of payments of a nation.

Macroeconomic Management is of 4 Types

1.Capitalism:

Capitalism is an economic system based on the private ownership of the means of production and their operation for profit.

2.Socialism:

Socialism is an economic system in which major industries are owned or regulated by the government, rather than by private businesses. It is different from capitalism, where private actors, like business owners and shareholders, can own the means of production for themselves.

3.mixed economic System:

A mixed economic system is a system that combines aspects of both capitalism and socialism. A mixed economic system protects private property and allows a level of economic freedom in the use of capital, but also allows for governments to interfere in economic activities in order to achieve social aims.

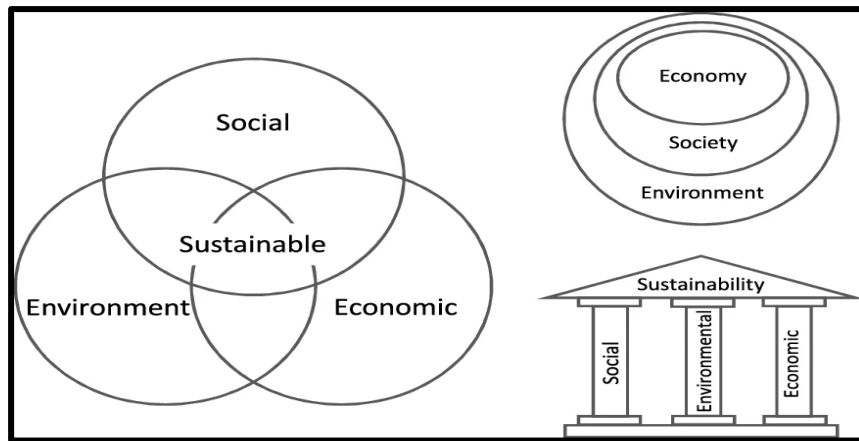
4.Communism:

Communism is a type of government as well as an economic system (a way of creating and sharing wealth). In a Communist system, individual people do not own land, factories, or machinery. Instead, the government or the whole community owns these things. Everyone is supposed to share the wealth that they create.

Sustainable development:

"Sustainable development is development that meets the needs of the present, without compromising the ability of future generations to meet their own needs."

- For sustainable development to be achieved, it is crucial to harmonize three core elements: **economic growth, social inclusion and environmental protection**. These elements are interconnected and all are crucial for the well-being of individuals and societies.
- The Brundtland Report in 1987 defined sustainable development as "development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs".
- The concept of sustainable development nowadays has a focus on economic development, social development and environmental protection for future generations.
- Sustainable development was first institutionalized with the Rio Process initiated at the 1992 Earth Summit in Rio de Janeiro.
- In 2015 the United Nations General Assembly (UNGA) adopted the Sustainable Development Goals (2015 to 2030) and explained how the goals are integrated and indivisible to achieve sustainable development at the global level.
- The UNGA's 17 goals address the global challenges, including poverty, inequality, climate change, environmental degradation, peace, and justice.



From reducing carbon emissions to promoting equality, the SDGs offer a framework for diverse stakeholders, including businesses, to make a positive impact on the world.

Comprising of 169 targets, these 17 SDGs include a collection of 17 interlinked objectives designed to serve as a shared blueprint for peace and prosperity of people and the planet - in the present and the future. These goals and their targets aim to accelerate actions that leave no one and no place behind.

In short, the 17 SDGs are:

Goal 1: No Poverty: End poverty in all its forms everywhere.

Goal 2: Zero Hunger: End hunger, achieve food security and improved nutrition and promote sustainable agriculture.

Goal 3: Good Health and Well-being: Ensure healthy lives and promote well-being for all at all ages.

Goal 4: Quality Education: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

Goal 5: Gender Equality: Achieve gender equality and empower all women and girls.

Goal 6: Clean Water and Sanitation: Ensure availability and sustainable management of water and sanitation for all.

Goal 7: Affordable and Clean Energy: Ensure access to affordable, reliable, sustainable and modern energy for all.

Goal 8: Decent Work and Economic Growth: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.

Goal 9: Industry, Innovation, and Infrastructure: Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation.

Goal 10: Reduced Inequality: Reduce inequality within and among countries.

Goal 11: Sustainable Cities and Communities: Make cities and human settlements inclusive, safe, resilient, and sustainable.

Goal 12: Responsible Consumption and Production: Ensure sustainable consumption and production patterns.

Goal 13: Climate Action: Take urgent action to combat climate change and its impacts.

Goal 14: Life Below Water: Conserve and sustainably use the oceans, seas, and marine resources for sustainable development.

Goal 15: Life on Land: Protect, restore, and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss.

Goal 16: Peace, Justice, and Strong Institutions: Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable, and inclusive institutions at all levels.

Goal 17: Partnerships to Achieve the Goal: Strengthen the means of implementation and revitalize the global partnership for sustainable development.

The 17 SDGs cut across various initiatives and each initiative often responds to various SDGs.

The benefits of using SDGs include:

1. Identify future business opportunities
2. Anticipate coming regulations
3. Gain new customers and retain existing customers
4. Attract talents and reduce turnover
5. Strengthen the company's reputation
6. Engage stakeholders
7. Answer requirements asked by investors, suppliers, and partners

At a more global level, using SDGs allow companies to have a common language to communicate about their impact and performance.

It also participates in stabilising societies and markets, as businesses cannot succeed in societies that fail.

Climate change Risk Rendition:

Climate change risk management approaches generally fall into four broad categories:

- 1) mitigation—efforts to reduce greenhouse gas emissions;
- 2) adaptation—increasing society's capacity to cope with changes in climate;
- 3) geoengineering or climate engineering—additional, deliberate manipulation of the earth system that is intended to counteract at least some of the impacts of greenhouse gas emissions; and

4) knowledge-based expansion—efforts to learn and understand more about the climate system, which can help support proactive risk management.

1.Mitigation:

By reducing emissions, mitigation reduces society's future contributions to greenhouse gas concentrations in the atmosphere. Ultimately, this can help reduce the amount that climate will change and thereby increase the potential that societal impacts will remain manageable. Approaches to reducing emissions fall into several categories.

These include

- 1) regulation;
- 2) research, development, and deployment of new technologies;
- 3) conservation;
- 4) efforts to increase public awareness;
- 5) positive incentives to encourage choices that lower emissions; and
- 6) adding a price to greenhouse gas emissions, which creates incentives to reduce emissions broadly.

2) Adaptation:

Adaptation involves planning for climate impacts, building resilience to those impacts, and improving society's capacity to respond and recover. This can help reduce damages and disruptions associated with climate change.

Adaptation policy can include regulation to decrease vulnerability (e.g., through land-use planning and building codes); response planning; disaster recovery; impact assessment for critical systems and resources (e.g., water, health, biological systems, agriculture, and infrastructure); observations and monitoring; and efforts to minimize compounding stresses such as traditional air pollution, habitat loss and degradation, invasive species, and nitrogen deposition.

3.geoengineering or climate engineering:

Geoengineering refers to deliberate, often global-scale, manipulations of the climate system. Two categories of geoengineering are most prevalent within scientific and policy discussions:

1. Solar radiation management (offsetting human-caused warming due to greenhouse gas emissions by reflecting incoming sunlight back to space) and
2. Carbon removal and sequestration (extracting carbon dioxide from the air and storing it deep in the ground or ocean).

Geoengineering could potentially help lower greenhouse gas concentrations in the atmosphere, counteract the warming influence of increasing greenhouse gas concentrations, address specific climate change impacts, or offer desperation strategies in the event that abrupt, catastrophic, or otherwise unacceptable climate change impacts become evident.

Geoengineering could also create new sources of risk because attempts to engineer the earth system on a large scale could lead to unintended and adverse consequences.

4.Expanding the knowledge base:

Allows policy makers to understand, select, and refine specific risk management strategies and to thereby increase the effectiveness of risk management efforts. Knowledge-base expansion can, in some cases, also reveal entirely new opportunities for protecting the climate system or reducing the risks of climate change impacts.

Financial Management of Disaster:

GFDRR strategy is driven by the central hypothesis that disaster risk is inherent to development, but that action can be taken to build resilience.

Research shows that the biggest driver of disaster losses globally is the growth of population and assets in hazard prone areas. Migration to

coastal areas and the expansion of cities in flood plains, coupled with inappropriate building practices, are among the main reasons for the

recent increase in disaster losses. DRM should therefore be integrated – or ‘mainstreamed’ – into all aspects of development, and the

outcomes systematically evaluated. Arresting the current trend of rising disaster losses – which is outpacing gains in prosperity – will require a

major shift in development planning and practice, including in the quality of the evidence to support such changes. This includes large-scale

investment in hard measures, like resilient infrastructure and the physical protection of cities. It will also require increased investment in soft

measures like early warning, land use planning, institution building and the application of new technologies and innovative solutions.

Five pillars of action provide the operational framework for GFDRR’s strategy. These are:

Pillar 1: Risk Identification - Risk assessments and risk communication

Pillar 2: Risk Reduction - Structural and non-structural measures; e.g. Infrastructure, land use planning, policies and regulation

Pillar 3: Preparedness - Early warning systems; support of emergency measures; contingency planning

Pillar 4: Financial Protection - Assessing and reducing contingent liabilities; budget appropriation and execution; ex-ante and ex-post

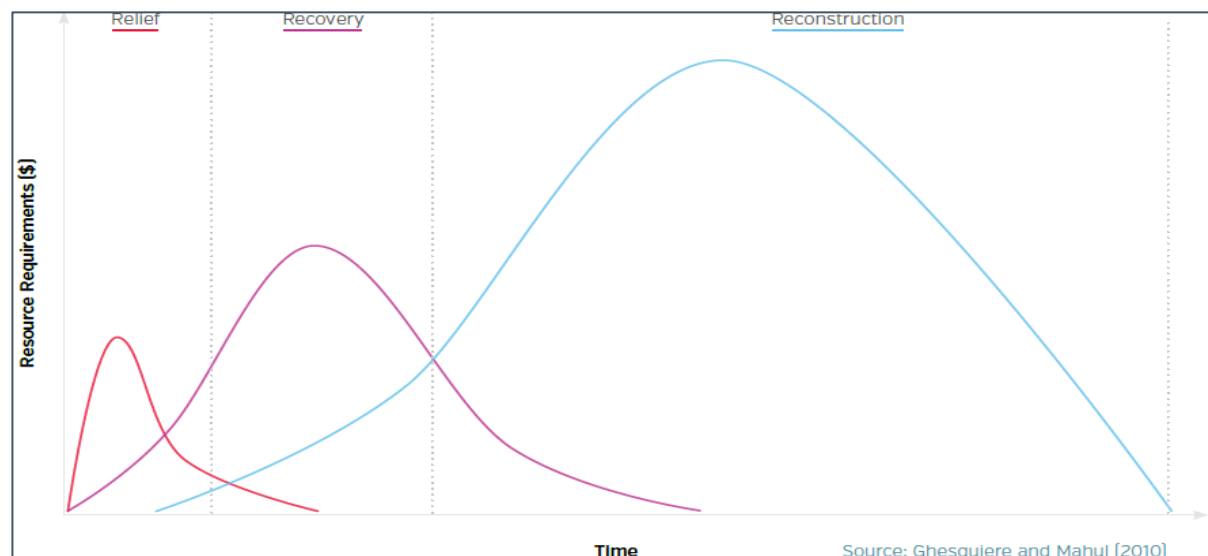
financing instruments

Pillar 5: Resilient Reconstruction - Resilient recovery and reconstruction policies; ex-ante design of institutional structures

Core Principles of Disaster Risk Finance

Governments seeking to evaluate and improve their financial resilience should be guided by four core principles. These principles do not tell decision makers what to do, but they provide a framework for evaluating policy decisions and financial instruments

1. Timeliness of funding: Speed matters but not all resources are needed at once. Understanding the timing of needs is essential. In the aftermath of a major disaster, the government will not require the money needed for the entire reconstruction program all at once. While immediate liquidity is crucial to support relief and early recovery operations, the government has more time to mobilize the larger resources for the reconstruction program. This variation in the timing of needs has clear implications for the design of cost-effective financial management of disasters.



2. Risk layering ensures that cheaper sources of money are used first and that the most expensive instruments are used only in exceptional circumstances. For example, insurance can provide cover against extreme events, but is not appropriate to protect against low-intensity events that recur regularly. To retain this lowest layer of risk, the government could consider setting up a dedicated contingency fund.

3. money reaches beneficiaries is as important as where it comes from. Governments require dedicated mechanisms and expertise to effectively allocate, disburse, and monitor recovery and reconstruction funds. Strong collaboration between the ministry of finance and the public entity tasked with spending post-disaster funds—such as local governments or agencies that maintain public infrastructure—is crucial. In addition, the disbursement system must balance policy makers' concern for fast disbursement with the transparency and accountability required by the public and donors

4.Data and analytics: To make sound financial decisions, governments need the right information.

Financial analysis of risk data and quantitative evidence empowers governments to take risk-informed decisions regarding their financial protection against disasters. Sound decision making requires actuarial analysis and tools to help governments understand and evaluate alternative financial instruments and strategies; user-friendly interfaces to bridge the gap between policy makers and underlying technical models; and quantitative analysis to leverage financial markets and private sector solutions.

Conclusion

The strong financial support during Natural disasters results in resilient strategies to fight back. It majorly covers and connects agriculture, urban development, social protection etc.